

SOFTWARE ENGINEERING

Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #54

Local Sports Tournament & Bracket Manager

Name: Guntupalli Manaswi

SRN: PES1UG24CS548

Semester: 5

Section: I

1. SYSTEM REQUIREMENTS SPECIFICATION DOCUMENT

Problem Statement: Local Sports Tournament & Bracket Manager (#54)

Actors: Team Captain; Tournament Director; Public Viewer

FUNCTIONAL REQUIREMENTS

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional - Bracket generation	The system shall automatically generate a seeded single-elimination or double-elimination bracket using the approved teams and their initial rankings.	High	Pass: In tests with 8 and 16 approved teams, every team appears once in the opening round, seed order is followed, and each winner or loser moves to the correct next slot for the selected format. Fail: Any team is missing, repeated, or sent to an invalid match path.	Bracket generation is the starting point of the tournament. Automating it reduces pairing mistakes and saves setup time.
FR-002	Functional - Team registration	The system shall allow a Team Captain to register a team by entering the team name, captain details, player roster, and initial ranking before registration closes.	High	Pass: A complete team record submitted before the deadline is saved and appears once in the Tournament Director participant list. Fail: The system accepts a duplicate team name, missing required details, or a late submission.	The Tournament Director needs complete team and ranking data before a fair bracket can be generated.
FR-003	Functional - Score recording	The system shall allow the Tournament Director to enter and confirm the final score of a scheduled match and shall advance the correct team to the next bracket slot.	High	Pass: A valid confirmed score is stored, the correct winner is identified, and that team appears in the expected next-round slot. Fail: An invalid score is accepted, a result is saved twice, or the wrong team advances.	Accurate score entry keeps the bracket current and avoids manual advancement errors.
FR-004	Functional - Standings and tiebreakers	The system shall recalculate wins, losses, points, goal difference, and configured tiebreakers whenever a match result is confirmed.	High	Pass: For a test set with at least four completed matches and one tie in points, all totals match the match records and tied teams are ordered using the configured rule. Fail: A total is incorrect or the tiebreaker order is not applied.	Consistent calculations prevent disputes and give every team the same basis for ranking.
FR-005	Functional - Publishing results	The system shall allow the Tournament Director to publish the latest bracket and standings so that Team Captains and Public Viewers can view the current tournament status.	Medium	Pass: After the Tournament Director selects Publish, a Team Captain and a Public Viewer can open the latest bracket and standings and see the same published version number and update time. Fail: Either viewer sees an older version or unpublished changes are displayed as final.	Publishing one approved version gives teams and spectators a reliable source of current tournament results.

NON-FUNCTIONAL REQUIREMENTS

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional - Performance	The system shall recalculate tournament standings, goal differentials, and leaderboard updates within 200 milliseconds after a valid score is submitted.	High	Pass: During a load test with 100 concurrent users, at least 95% of 1,000 valid score submissions complete the standings, goal-differential, and leaderboard recalculation within 200 ms. Fail: Fewer than 95% of the submissions meet the 200 ms limit.	Fast recalculation keeps standings current when matches are close together and many users are checking results.
NFR-002	Nonfunctional - Security	The system shall use role-based access control so that only an authenticated Tournament Director can create tournaments, change scores, generate brackets, or publish standings, and each successful Director action shall be recorded in an audit log.	High	Pass: Team Captains, Public Viewers, and signed-out users are denied all Director-only actions, while every successful Tournament Director action records the user, action, and timestamp. Fail: An unauthorized Director-only action succeeds or a required audit entry is missing.	Scores and brackets affect the whole competition, so administrative changes must be authorized and traceable.